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FOR JEE MAIN EXAM PREPARATION

# JEE Main 2026 Question Paper · B Tech

| EXAM | YEAR | TYPE | SUBJECT | DATE | SHIFT |
|---|---|---|---|---|---|
| JEE Main | 2026 | Question Paper | B Tech | 6th Apr | Shift 1 |

# National Testing Agency

| | |
|---|---|
| **Question Paper Name :** | B Tech 6th Apr 2026 Shift 1 |
| **Subject Name :** | B. Tech |
| **Creation Date :** | 2026-04-06 14:36:27 |
| **Duration :** | 180 |
| **Total Marks :** | 300 |
| **Display Marks:** | Yes |

## B. Tech

| | |
|---|---|
| **Group Number :** | 1 |
| **Group Id :** | 69527828 |
| **Group Maximum Duration :** | 0 |
| **Group Minimum Duration :** | 180 |
| **Show Attended Group? :** | No |
| **Edit Attended Group? :** | No |
| **Break time :** | 0 |
| **Group Marks :** | 300 |

## Mathematics Section A

| | |
|---|---|
| **Section Id :** | 695278151 |
| **Section Number :** | 1 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Sub-Section Id :** | 695278151 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 1 Question Id : 6952782136 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x)=\sin^{-1}\left(\frac{x+[x]}{3}\right)$ is $[\alpha, \beta)$, then $\alpha^2+\beta^2$ is equal to:

**Options :**

6952787496. 2

6952787497. 5

6952787498. 10

6952787499. 13

**Question Number : 2 Question Id : 6952782137 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let one root of the quadratic equation in $x$:
$(k^2 - 15k + 27)x^2 + 9(k - 1)x + 18 = 0$
be twice the other. Then the length of the latus rectum of the parabola $y^2 = 6kx$ is equal to:

**Options :**

6952787500. $4$

6952787501. $6$

6952787502. $8$

6952787503. $12$

**Question Number : 3 Question Id : 6952782138 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $e_1$ and $e_2$ be two distinct roots of the equation $x^2 - ax + 2 = 0$. Let the sets
$\{a \in \mathbb{R} : e_1$ and $e_2$ are the eccentricities of hyperbolas$\} = (\alpha, \beta)$, and
$\{a \in \mathbb{R} : e_1$ and $e_2$ are the eccentricities of an ellipse and a hyperbola, respectively$\} = (\gamma, \infty)$.
Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to:

**Options :**

6952787504. $18$

6952787505. $22$

6952787506. $26$

6952787507. $34$

**Question Number : 4 Question Id : 6952782139 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let the set of all values of $k \in \mathbb{R}$ such that the equation $z(\bar{z} + 2 + i) + k(2 + 3i) = 0$, $z \in \mathbb{C}$, has at least one solution, be the interval $[\alpha, \beta]$. Then $9(\alpha + \beta)$ is equal to:

**Options :**

6952787508. $-10$

6952787509. $-8$

6952787510. $10\sqrt{13}$

6952787511. $8\sqrt{13}$

**Question Number : 5 Question Id : 6952782140 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The value of $1^3 - 2^3 + 3^3 - \ldots + 15^3$ is:

**Options :**

6952787512. $1706$

6952787513. 1856

6952787514. 1982

6952787515. 2403

**Question Number : 6 Question Id : 6952782141 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8. If the first term of the A.P. is equal to the common ratio of the G.P. and the first term of the G.P. is equal to common difference of the A.P., then the sum of all possible values of the first term of the G.P. is:

**Options :**

6952787516. $\frac{34}{9}$

6952787517. $\frac{34}{13}$

6952787518. $\frac{32}{9}$

6952787519. $\frac{32}{13}$

**Question Number : 7 Question Id : 6952782142 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The number of 4-letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters of the word INCONSEQUENTIAL, without repeating any letter, is:

**Options :**

6952787520. 2670

6952787521. 2840

6952787522. 2920

6952787523. 3600

**Question Number : 8 Question Id : 6952782143 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If the coefficients of the middle terms in the binomial expansions of $(1 + \alpha x)^{26}$ and $(1 - \alpha x)^{28}$, $\alpha \neq 0$, are equal, then the value of $\alpha$ is:

**Options :**

6952787524. 1

6952787525. $\frac{14}{13}$

6952787526. $\frac{27}{7}$

6952787527. $\frac{7}{27}$

**Question Number : 9 Question Id : 6952782144 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A data consists of 20 observations $x_1, x_2, ..., x_{20}$. If $\sum_{i=1}^{20}(x_i+5)^2 = 2500$ and $\sum_{i=1}^{20}(x_i-5)^2 = 100$, then the ratio of mean to standard deviation of this data is:

**Options :**

6952787528. 2:1

6952787529. 3:1

6952787530. 3:2

6952787531. 4:1

**Question Number : 10 Question Id : 6952782145 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A bag contains (N + 1) coins − N fair coins, and one coin with 'Head' on both sides. A coin is selected at random and tossed. If the probability of getting 'Head' is $\frac{9}{16}$, then N is equal to:

**Options :**

6952787532. 5

6952787533. 7

6952787534. 8

6952787535. 9

**Question Number : 11 Question Id : 6952782146 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If the eccentricity $e$ of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, passing through $\left(6, 4\sqrt{3}\right)$, satisfies $15(e^2 + 1) = 34e$, then the length of the latus rectum of the hyperbola $\frac{x^2}{b^2}-\frac{y^2}{2\left(a^2+1\right)}=1$ is:

**Options :**

6952787536. 10

6952787537. 20

6952787538. 25

6952787539. 30

**Question Number : 12 Question Id : 6952782147 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let chord PQ of length $3\sqrt{13}$ of the parabola $y^2 = 12x$ be such that the ordinates of points P and Q are in the ratio 1:2. If the chord PQ subtends an angle $\alpha$ at the focus of the parabola, then $\sin\alpha$ is equal to:

**Options :**

6952787540. $\frac{3}{5}$

6952787541. $\frac{4}{5}$

6952787542. $\frac{5}{13}$

6952787543. $\frac{12}{13}$

**Question Number : 13 Question Id : 6952782148 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $0 < \alpha < 1, \beta = \frac{1}{3\alpha}$ and $\tan^{-1}(1-\alpha) + \tan^{-1}(1-\beta) = \frac{\pi}{4}$. Then $6(\alpha + \beta)$ is equal to:

**Options :**

6952787544. 6

6952787545. 7

6952787546. 8

6952787547. 9

**Question Number : 14 Question Id : 6952782149 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $S = \left\{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\right\}$.

Then $\sum_{\theta \in S} \theta$ is equal to:

**Options :**

6952787548. $-\frac{2\pi}{3}$

6952787549. $-\frac{4\pi}{3}$

6952787550. $\frac{2\pi}{3}$

6952787551. $\frac{4\pi}{3}$

**Question Number : 15 Question Id : 6952782150 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let the image of the point P(1, 6, $a$) in the line $L: \frac{x}{1} = \frac{y-1}{2} = \frac{z-a+1}{b}, b > 0,$ be $\left(\frac{a}{3}, 0, a+c\right)$. If S($\alpha$, $\beta$, $\gamma$), $\alpha > 0$, is the point on L such that the distance of S from the foot of perpendicular from the point P on L is $2\sqrt{14}$, then $\alpha + \beta + \gamma$ is equal to:

**Options :**

6952787552. 19

6952787553. 20

6952787554. 21

6952787555. 22

**Question Number : 16 Question Id : 6952782151 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let a line L be perpendicular to both the lines $L_1: \frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $L_2: \frac{x-2}{1} = \frac{y-4}{4} = \frac{z-6}{7}$.

If $\theta$ is the acute angle between the lines L and $L_3: \frac{x-\frac{8}{7}}{2} = \frac{y-\frac{4}{7}}{1} = \frac{z}{2}$, then $\tan\theta$ is equal to:

**Options :**

6952787556. $\frac{3}{2}\sqrt{2}$

6952787557. $\frac{5}{2}\sqrt{2}$

6952787558. $\frac{5}{3}\sqrt{2}$

6952787559. $\frac{4}{3}\sqrt{2}$

**Question Number : 17 Question Id : 6952782152 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The value of $\lim\limits_{x\to 0}\left(\frac{x^2\sin^2 x}{x^2 - \sin^2 x}\right)$ is:

**Options :**

6952787560. 2

6952787561. 3

6952787562. 4

6952787563. 6

**Question Number : 18 Question Id : 6952782153 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left( \frac{32\cos^4 x}{1+e^{\sin x}} \right) dx$ is:

**Options :**

6952787564. $4\pi + 2$

6952787565. $3\pi + 8$

6952787566. $3\pi + 4$

6952787567. $4\pi + 3$

**Question Number : 19 Question Id : 6952782154 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The area of the region $\{(x, y) : 0 \le y \le 6 - x, y^2 \ge 4x - 3, x \ge 0\}$ is:

**Options :**

6952787568. 8

6952787569. 9

6952787570. 12

6952787571. 15

**Question Number : 20 Question Id : 6952782155 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $e$ be the base of natural logarithm and let $f : \{1, 2, 3, 4\} \to \{1, e, e^2, e^3\}$ and $g : \{1, e, e^2, e^3\} \to \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}\right\}$ be two bijective functions such that $f$ is strictly decreasing and g is strictly increasing. If $\phi(x) = \left[ f^{-1}\left\{ g^{-1}\left(\frac{1}{2}\right)\right\}\right]^x$, then the area of the region R = $\{(x, y): x^2 \le y \le \phi(x), 0 \le x \le 1\}$ is:

**Options :**

6952787572. $\frac{3 - \log_e(2)}{3\log_e(2)}$

6952787573. $\frac{1}{3\log_e(2)}$

6952787574. $3 + \log_e(2)$

6952787575. $\frac{3 + \log_e(2)}{2 + \log_e(3)}$

# Mathematics Section B

| | |
|---|---|
| **Section Id :** | 695278152 |
| **Section Number :** | 2 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 5 |
| **Number of Questions to be attempted :** | 5 |
| **Section Marks :** | 20 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Sub-Section Id :** | 695278152 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 21 Question Id : 6952782156 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Let $A=\begin{bmatrix}-1 & 1 & -1\\ 1 & 0 & 1\\ 0 & 0 & 1\end{bmatrix}$ satisfy

$$A^2+\alpha\left(adj\left(adj(A)\right)\right)+\beta\left(adj(A)\left(adj\left(adj(A)\right)\right)\right)=\begin{bmatrix}2 & -2 & 2\\ -2 & 0 & -1\\ 0 & 0 & -1\end{bmatrix}$$ for

some $\alpha, \beta \in \mathbb{R}$.

Then $(\alpha-\beta)^2$ is equal to _____

**Response Type :** Numeric

**Evaluation Required For SA :** Yes

**Show Word Count :** Yes

**Answers Type :** Equal

**Text Areas :** PlainText

**Possible Answers :**

1

**Question Number : 22 Question Id : 6952782157 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Let the centre of the circle $x^2+y^2+2gx+2fy+25=0$ be in the first quadrant and lie on the line $2x-y=4$. Let the area of an equilateral triangle inscribed in the circle be $27\sqrt{3}$. Then the square of the length of the chord of the circle on the line $x=1$ is_____.

**Response Type :** Numeric

**Evaluation Required For SA :** Yes

**Show Word Count :** Yes

**Answers Type :** Equal

**Text Areas :** PlainText

**Possible Answers :**

1

**Question Number : 23 Question Id : 6952782158 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{j}-\hat{k}$ and $\vec{c}$ be three vectors such that $\vec{a}\times\vec{c}=\vec{b}$ and $\vec{a}\cdot\vec{c}=3$, then $\vec{c}\cdot\left(\vec{a}-2\vec{b}\right)$ is equal to _____.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 24 Question Id : 6952782159 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

For the functions $f(\theta) = \alpha \tan^2\theta + \beta \cot^2\theta$, and

$g(\theta) = \alpha \sin^2\theta + \beta \cos^2\theta$, $\alpha > \beta > 0$, let $\min\limits_{0<\theta<\frac{\pi}{2}} f(\theta) = \max\limits_{0<\theta<\pi} g(\theta)$. If the first term of a G.P. is $\left(\frac{\alpha}{2\beta}\right)$, its common ratio is $\left(\frac{2\beta}{\alpha}\right)$ and the sum of its first 10 terms is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to _____.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 25 Question Id : 6952782160 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Let $y = y(x)$ be the solution of the differential equation $\left(x^2 - x\sqrt{x^2-1}\right)dy + \left(y\left(x - \sqrt{x^2-1}\right) - x\right)dx = 0$, $x \geq 1$. If $y(1)=1$, then the greatest integer less than $y\left(\sqrt{5}\right)$ is _____.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

# Physics Section A

| | |
|---|---|
| **Section Id :** | 695278153 |
| **Section Number :** | 3 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Sub-Section Id :** | 695278153 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 26 Question Id : 6952782161 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The density $\rho$ of a uniform cylinder is determined by measuring its mass $m$, length $l$ and diameter $d$. The measured values of $m$, $l$ and $d$ are $97.42 \pm 0.02$ g, $8.35 \pm 0.05$ mm and $20.20 \pm 0.02$ mm, respectively. Calculated percentage fractional error in $\rho$ is _____.

**Options :**

6952787581. 0.63%

6952787582. 0.82%

6952787583. 0.72%

6952787584. 0.25%

**Question Number : 27 Question Id : 6952782162 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The potential energy of a particle changes with distance $x$ from a fixed origin as $V = \frac{A\sqrt{x}}{x+B}$, where $A$ and $B$ are constant with appropriate dimensions. The dimensions of $AB$ are_____.

**Options :**

6952787585. $[M^1 L^{5/2} T^{-2}]$

6952787586. $[M^{3/2} L^{5/2} T^{-2}]$

6952787587. $[M^1 L^2 T^{-2}]$

6952787588. $[M^1 L^{7/2} T^{-2}]$

**Question Number : 28 Question Id : 6952782163 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The rain drop of mass 1 g, starts with zero velocity from a height of 1 km. It hits the ground with a speed of 5 m/s. The work done by the unknown resistive force is _____J.
(take g = 10 m/s$^2$)

**Options :**

6952787589. $-8.75$

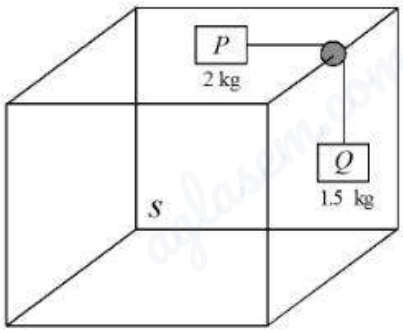
6952787590. $-8.35$

6952787591. $-9.55$

6952787592. $-9.98$

**Question Number : 29 Question Id : 6952782164 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Two blocks ($P$ and $Q$) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless pully which is fixed on the edge of a cube ($S$), as shown in the figure below. Block $P$ is positioned on the top surface which has no friction and block $Q$ is in contact with side-surface, having coefficient friction $\mu$. The cube ($S$) moves towards the right with acceleration of $\frac{g}{2}$, where $g$ is gravitational acceleration. During this movement the block $P$ and $Q$ remain stationary. The value of $\mu$ is _____.
(take g = 10 m/s$^2$)




**Options :**

6952787593. 0.33

6952787594. 0.67

6952787595. 1

6952787596. 0.5

**Question Number : 30 Question Id : 6952782165 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A lift of mass 1600 kg is supported by thick iron wire. If the maximum stress which the wire can withstand is $4 \times 10^8$ N/m$^2$ and its radius is 4 mm, then maximum acceleration the lift can take is _____ m/s$^2$.
(take g = 10 m/s$^2$ and $\pi = 3.14$)

**Options :**

6952787597. 2.56

6952787598. 3.89

6952787599. 4.32

6952787600. 5.16

**Question Number : 31 Question Id : 6952782166 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A solid sphere of radius 4 cm and mass 5 kg is rotating (rotation axis is passing through the centre of the sphere) with an angular velocity of 1200 rpm. It is brought to rest in 10 s by applying a constant torque. The torque applied and the number of rotations it made before it comes to rest are _____ and _____ respectively.

**Options :**

6952787601. $0.128\ \pi$ Nm, 100

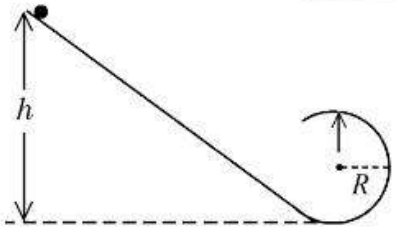
6952787602. $0.0128\ \pi$ Nm, 50

6952787603. $0.128\ \pi$ Nm, 50

6952787604. $0.0128\ \pi$ Nm, 100

**Question Number : 32 Question Id : 6952782167 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A smooth inclined plane ends in a vertical circular loop, as shown in the figure. A small body is released from height $h$ as shown. If the body exerts a force of three times its weight on the plane at the highest point of circle then the height $h = \alpha R$. The value of $\alpha$ is _____.




**Options :**

6952787605. 2

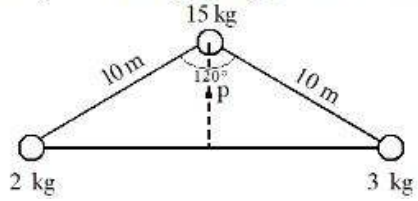
6952787606. 4

6952787607. 3

6952787608. 6

**Question Number : 33 Question Id : 6952782168 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The position of center of mass of three masses 2 kg, 3 kg and 15 kg placed with respect to mid point ($p$) of normal bisector, as shown in the figure is_____.




**Options :**

6952787609. $\left(\frac{\sqrt{3}}{4}, 1.25\right)$

6952787610. $\left(\frac{\sqrt{3}}{4}, 1.0\right)$

6952787611. $(0,0)$

6952787612. $(1.25,0)$

**Question Number : 34 Question Id : 6952782169 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The two wires $A$ and $B$ of equal cross-section but of different materials are joined together. The ratio of Young's modulus of wire $A$ and wire $B$ is 20/11. When the joined wire is kept under certain tension the elongations in the wires $A$ and $B$ are equal. If the length of wire $A$ is 2.2 m, then the length of wire $B$ is _____m.

**Options :**

6952787613. 1.1

6952787614. 2.22

6952787615. 1.21

6952787616. 4.44

**Question Number : 35 Question Id : 6952782170 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 400 K. Keeping the temperature of one vessel constant at 400 K the second vessel temperature is raised to 500 K. The final pressure in the vessels is _____kPa.

**Options :**

6952787617. 100

6952787618. 120

6952787619. 90

6952787620. 105

**Question Number : 36 Question Id : 6952782171 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

In interference experiment the path difference between two interfering waves at a point $A$ on the screen is $\lambda/3$, where $\lambda$ is the wavelength of these waves, and at another point $B$ the path difference is $\lambda/6$. The ratio of intensities at points $A$ and $B$ is_____.

**Options :**

6952787621. 3

6952787622. 4

6952787623. 1/3

6952787624. 1/4

**Question Number : 37 Question Id : 6952782172 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A particle is executing simple harmonic motion. Its amplitude is $A$ and time period is 5 sec. The time required by it to move from $x = A$ to $x = \frac{A}{\sqrt{2}}$ is _____ sec.

**Options :**

6952787625. 1/4

6952787626. 5/4

6952787627. 5/8

6952787628. 3/8

**Question Number : 38 Question Id : 6952782173 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A thin half ring of radius 35 cm is uniformly charged with a total charge of $Q$ coulomb. If the magnitude of the electric field at centre of the half ring is 100 V/m, then the value of $Q$ is _____ nC.

($\epsilon_o = 8.85 \times 10^{-12}$ C$^2$/Nm$^2$ and $\pi = 3.14$)

**Options :**

6952787629. 2.14

6952787630. 2.44

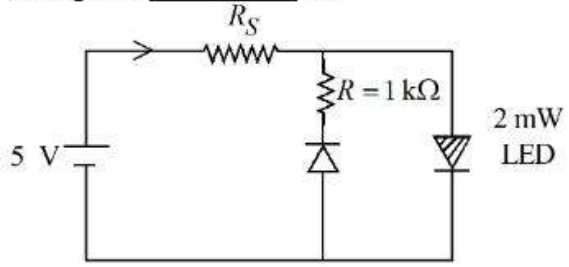
6952787631. 3.25

6952787632. 0.7

**Question Number : 39 Question Id : 6952782174 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The maximum rated power of the LED is 2 mW and it is used in the circuit with input voltage of 5 V as shown in the figure below. The current through resistance $R_S$ is 0.5 mA.

The minimum value of the resistance of $R_S$, to ensure that the LED is not damaged is _____ kΩ.




**Options :**

6952787633. 6

6952787634. 2

6952787635. 4

6952787636. 5

**Question Number : 40 Question Id : 6952782175 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A point light source emits E.M. waves in free space. A detector, placed at a distance of $L$ m, measures the intensity as $I_o$. The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45°. The measured intensity at new location will be _____.

**Options :**

6952787637. $\frac{I_o}{4}$

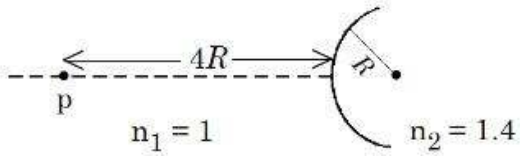
6952787638. $I_o$

6952787639. $\frac{I_o}{\sqrt{2}}$

6952787640. $\frac{I_o}{2}$

**Question Number : 41 Question Id : 6952782176 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A spherical interface lens of radius $R$ separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A point source is placed at a distance of $4R$ in front of spherical interface. The magnitude of the magnification of point source image is _____.




**Options :**

6952787641. 1.66

6952787642. 2.33

6952787643. 2.66

6952787644. 1.33

**Question Number : 42 Question Id : 6952782177 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A small cube of side 1 mm is placed at the centre of a circular loop of radius 10 cm carrying a current of 2 A. The magnetic energy stored inside the cube is $\alpha \times 10^{-14}$ J. The value of $\alpha$ is _____.
($\mu_o = 4\pi \times 10^{-7}$ Tm/A, $\pi = 3.14$)

**Options :**

6952787645. 6.28

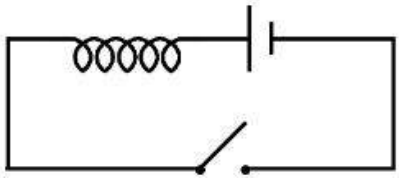
6952787646. $6.28 \times 10^{-6}$

6952787647. 628

6952787648. $6.28 \times 10^{-4}$

**Question Number : 43 Question Id : 6952782178 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

An inductor of inductance 10 mH having resistance of 100 Ω is connected to battery of E.M.F. 1.0 V through a switch as shown in the figure below. After switch is closed, the ratio of instantaneous voltages across the inductor when the current passing through it is 2 mA and 4 mA is _____.



**Options :**

6952787649. 4/3

6952787650. 3/4

6952787651. 5/3

6952787652. 3/5

**Question Number : 44 Question Id : 6952782179 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The ratio of momentum of the photons of the $1^{st}$ and $2^{nd}$ line of Balmer series of Hydrogen atoms is α/β. The possible values of α and β are:-

**Options :**

6952787653. 27 and 20

6952787654. 3 and 16

6952787655. 5 and 36

6952787656. 20 and 27

**Question Number : 45 Question Id : 6952782180 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A LCR series circuit driven with $E_{rms} = 90$ V at frequency $f_d = 30$ Hz has resistance $R = 80$ Ω, an inductance with inductive reactance $X_L = 20.0$ Ω and capacitance with capacitive reactance $X_C = 80.0$ Ω. The power factor of the circuit is_____.

**Options :**

6952787657. 0.8

6952787658. 0.64

6952787659. 0.9

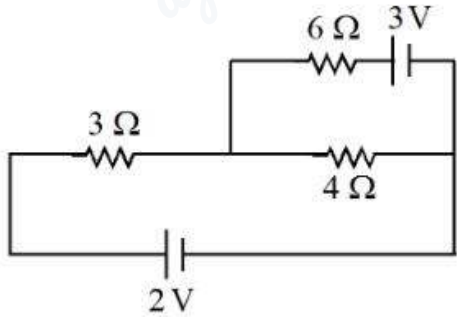
6952787660. 0.5

## Physics Section B

| | |
|---|---|
| **Section Id :** | 695278154 |
| **Section Number :** | 4 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 5 |
| **Number of Questions to be attempted :** | 5 |
| **Section Marks :** | 20 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Sub-Section Id :** | 695278154 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 46 Question Id : 6952782181 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Refer to the circuit diagram given below. The heat generated across the 6 Ω resistance in 100 second is $\frac{\alpha}{100}$ J . The value of α is _____. (Nearest integer)




**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 47 Question Id : 6952782182 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

An unpolarized light of intensity $I_0$ passes through polarizer and then through a certain optically active solution and finally it goes to analyser. If the angle between analyser and polariser is 0° and intensity of light emerged from analyser is $\frac{3}{8}I_0$, the angle of rotation of the light by the solution with respect to analyser is _____ degrees.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 48 Question Id : 6952782183 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

The energy released when $\frac{7}{17.13}$ kg of $^{7}_{3}Li$ is converted into $^{4}_{2}He$ by proton bombardment is $\alpha \times 10^{32}$ eV. The value of $\alpha$ is ____. (Nearest integer)
(Mass of $^{7}_{3}Li$ = 7.0183 u, mass of $^{4}_{2}He$ = 4.004 u, mass of proton = 1.008 u and 1 u = 931 MeV/$c^2$ and Avogadro number = $6.0 \times 10^{23}$)

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 49 Question Id : 6952782184 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

A three coulomb charge moves from the point (0, −2, −5) to the point (5, 1, 2) in an electric field expressed as $\vec{E} = 2x\hat{i} + 3y^2\hat{j} + 4\hat{k}$ N/C. The work done in moving the charge is _____J.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 50 Question Id : 6952782185 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

A certain gas is isothermally compressed to $\left(\frac{1}{3}\right)^{rd}$ of its initial volume ($V_o$ = 3 litre) by applying required pressure. If the bulk modulus of the gas is $3 \times 10^5$ N/m$^2$, the magnitude of work done on the gas is _____J.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

# Chemistry Section A

| | |
|---|---|
| **Section Id :** | 695278155 |
| **Section Number :** | 5 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Sub-Section Id :** | 695278155 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 51 Question Id : 6952782186 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

An oxide of iron contains 69.9% iron, its empirical formula, is:

(Given : Molar mass of Fe and O are 56 and 16 g $mol^{-1}$ respectively.)

**Options :**

6952787666. $FeO$

6952787667. $Fe_2O_3$

6952787668. $Fe_3O_4$

6952787669. $FeO_3$

**Question Number : 52 Question Id : 6952782187 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If shortest wavelength of hydrogen atom in Lyman series is $x$, then longest wavelength in Balmer series of $He^+$ is:

**Options :**

6952787670. $\frac{9x}{5}$

6952787671. $\frac{36x}{5}$

6952787672. $\frac{x}{4}$

6952787673. $\frac{5x}{9}$

**Question Number : 53 Question Id : 6952782188 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match the **LIST-I** with **LIST-II**

| | **List-I Orbital** | | **List-II Radial nodes and nodal plane** |
|---|---|---|---|
| A. | 2s | I. | 1 Radial node + two nodal planes |
| B. | 3s | II. | 1 Radial node + one nodal plane |
| C. | 3p | III. | 2 Radial nodes + No nodal plane |
| D. | 4d | IV. | 1 Radial node + No nodal plane |

Choose the ***correct*** answer from the options given below:

**Options :**

6952787674. A-IV, B-I, C-III, D-II

6952787675. A-IV, B-II, C-III, D-I

6952787676. A-III, B-I, C-IV, D-II

6952787677. A-IV, B-III, C-II, D-I

**Question Number : 54 Question Id : 6952782189 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The pairs among
$A = \left[ SO_3^{2-}, CO_3^{2-} \right]$, $B = \left[ O_2^{2-}, F_2 \right]$, $C = \left[ CN^-, CO \right]$, $D = \left[ NH_3, H_3O^+ \right]$ and
$E = \left[ MnO_4^{2-}, CrO_4^{2-} \right]$ that do not have similar Lewis dot structure are

**Options :**

6952787678. A, B and E

6952787679. A and E

6952787680. B, C and D

6952787681. C and D

**Question Number : 55 Question Id : 6952782190 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Arrange the following isothermal processes in order of the magnitude of the work (p − V) involved between states 1 and 2.

A. Expansion in single stage $w_A$
B. Expansion in multi stages $w_B$
C. Compression in single stage $w_C$
D. Compression in multi stages $w_D$

Choose the correct option.

**Options :**

6952787682. $|w_B| > |w_A| > |w_C| > |w_D|$

6952787683. $|w_C| > |w_D| > |w_A| > |w_B|$

6952787684. $|w_C| > |w_D| > |w_B| > |w_A|$

6952787685. $|w_B| > |w_A| > |w_D| > |w_C|$

**Question Number : 56 Question Id : 6952782191 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

When 0.25 moles of a non-volatile, non-ionizable solute was dissolved in 1 mole of a solvent the vapor pressure of solution was $x$ % of vapor pressure of pure solvent. What is $x$ %?

**Options :**

6952787686. 50%

6952787687. 60%

6952787688. 70%

6952787689. 80%

**Question Number : 57 Question Id : 6952782192 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

One mole each of He and A(g) are taken in a 10 L closed flask and heated to 400 K to establish the following equilibrium.

$A(g) \rightleftharpoons B(g)$

$K_c$ for this reaction at 400 K is 4.0. The partial pressures (in atm) of He and B(g) are respectively (at equilibrium)

(Assume He, A(g) and B(g) behave as ideal gases)

(Given : R = 0.082 L atm $K^{-1}$ $mol^{-1}$)

**Options :**

6952787690. 3.28, 2.624

6952787691. 2.624, 3.28

6952787692. 3.28, 0.656

6952787693. 0.656, 6.56

**Question Number : 58 Question Id : 6952782193 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Consider the following data.

| Electrolyte | $\Lambda^\circ_m$ (S $cm^2$ $mol^{-1}$) |
|---|---|
| $BaCl_2$ | $x_1$ |
| $H_2SO_4$ | $x_2$ |
| HCl | $x_3$ |

$BaSO_4$ is sparingly soluble in water. If the conductivity of the saturated $BaSO_4$ solution is $x$ S $cm^{-1}$ then the solubility product of $BaSO_4$ can be given as

(Here $\Lambda_m = \Lambda^\circ_m$)

**Options :**

6952787694. $\frac{10^6 x^2}{\alpha^2 \left(x_1 + x_2 - 2x_3\right)^2}$

6952787695. $\frac{x^2}{\left(x_1 + x_2 - 2x_3\right)^2}$

6952787696. $\frac{\alpha^2 \left(x_1 + x_2 - 2x_3\right)^2}{10^6 x^2}$

6952787697. $\frac{x^2}{\left(x_1 + x_2 + 2x_3\right)^2}$

**Question Number : 59 Question Id : 6952782194 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements:

**Statement I:** Aluminium is more electropositive than thallium as the standard electrode potential value of $E^\circ_{Al^{3+}/Al}$ is negative and $E^\circ_{Tl^{3+}/Tl}$ is positive.

**Statement II:** The sum of first three ionization enthalpies of boron is very high when compared to that of aluminium. Due to this reason boron forms covalent compounds only and aluminium forms $Al^{3+}$ ion.

In the light of the above statements, choose the ***correct*** answer from the options given below

**Options :**

6952787698. Both Statement I and Statement II are true

6952787699. Both Statement I and Statement II are false

6952787700. Statement I is true but Statement II is false

6952787701. Statement I is false but Statement II is true

**Question Number : 60 Question Id : 6952782195 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The **correct** statements among the following are.

A. Basic vanadium oxide is used in the manufacture of $H_2SO_4$.
B. The spin-only magnetic moment value of the transition metal halide employed in Ziegler-Natta polymerization is 2.84 BM.
C. The p-block metal compound employed in Ziegler-Natta polymerization has the metal in +3 oxidation state.
D. The number of electrons present in the outer most 'd' orbital of metal halide employed in Wacker process is 8.

Choose the correct answer from the options given below:

**Options :**

6952787702. A and B Only

6952787703. A, C and D Only

6952787704. C and D Only

6952787705. B, C and D Only

**Question Number : 61 Question Id : 6952782196 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match the **LIST-I** with **LIST-II**

| List-I Electronic configuration of tetrahedral metal ion | | List-II Crystal Field Stabilization Energy ($\Delta_t$) | |
|---|---|---|---|
| A. | $d^2$ | I. | − 0.6 |
| B. | $d^4$ | II. | − 0.8 |
| C. | $d^6$ | III. | − 1.2 |
| D. | $d^8$ | IV. | − 0.4 |

Choose the ***correct*** answer from the options given below:

**Options :**

6952787706. A-III, B-IV, C-II, D-I

6952787707. A-III, B-I, C-IV, D-II

6952787708. A-III, B-IV, C-I, D-II

6952787709. A-II, B-I, C-IV, D-III

**Question Number : 62 Question Id : 6952782197 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Which of the following are true about the energy of the given d-orbitals of a tetrahedral complex?

A. $d_{xy} = d_{xz} > d_{x^2-y^2}$
B. $d_{xy} = d_{yz} > d_{z^2}$
C. $d_{x^2-y^2} > d_{z^2} > d_{xz}$
D. $d_{x^2-y^2} = d_{z^2} < d_{xz}$

Choose the correct answer from the given below:

**Options :**

6952787710. A, B and D only

6952787711. A and B only

6952787712. B and D only

6952787713. B, C and D only

**Question Number : 63 Question Id : 6952782198 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

$R_f$ value for 2-methylpropene in a solvent system (Ethyl acetate + ether) is 0.42.
2-methylpropene is treated with dilute $H_2SO_4$ to give major organic product (X).
$R_f$ value for (X) in the same solvent system under identical condition will be:

**Options :**

6952787714. 0.42

6952787715. 0.82

6952787716. 0.62

6952787717. 0.12

**Question Number : 64 Question Id : 6952782199 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements:

**Statement I:** 2,6-diethylcyclohexanone and 6-methyl-2-n-propylcyclohexanone are metamers.

**Statement II:** 2,2,6,6 - tetramethylcyclohexanone exhibits keto-enol tautomerism.

In the light of the above statements, choose the ***correct*** answer from the options given below

**Options :**

6952787718. Both Statement I and Statement II are true

6952787719. Both Statement I and Statement II are false

6952787720. Statement I is true but Statement II is false

6952787721. Statement I is false but Statement II is true

**Question Number : 65 Question Id : 6952782200 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements:

**Statement I:** Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of $CH_3MgBr$ with water.

**Statement II:** Methane cannot be prepared from unsaturated hydrocarbons and by Wurtz reaction.

In the light of the above statements, choose the ***correct*** answer from the options given below

**Options :**

6952787722. Both Statement I and Statement II are true

6952787723. Both Statement I and Statement II are false

6952787724. Statement I is true but Statement II is false

6952787725. Statement I is false but Statement II is true

**Question Number : 66 Question Id : 6952782201 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements:

**Statement I:** 3-phenylpropene reacts with HBr and gives secondary alkyl bromide having a chiral carbon atom as the major product.

**Statement II:** Aryl chlorides and aryl cyanides can be prepared by Sandmeyer reaction as well as Gattermann reaction.

In the light of the above statements, choose the ***correct*** answer from the options given below

**Options :**

6952787726. Both Statement I and Statement II are true

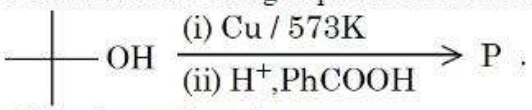
6952787727. Both Statement I and Statement II are false

6952787728. Statement I is true but Statement II is false

6952787729. Statement I is false but Statement II is true

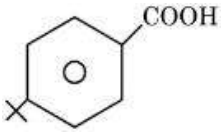
**Question Number : 67 Question Id : 6952782202 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

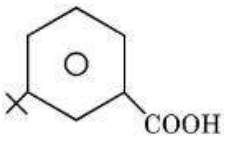
Consider the following sequence of reactions

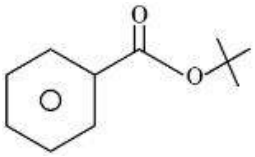


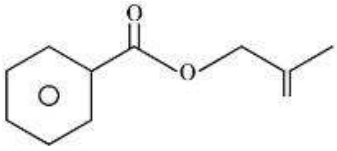

The major product P is:

**Options :**

6952787730. 


6952787731. 


6952787732. 


6952787733. 


**Question Number : 68 Question Id : 6952782203 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Arrange the following compounds according to increasing order of boiling points.

n - $C_4H_9OH$ (A), n - $C_4H_9NH_2$ (B), n - $C_4H_{10}$ (C) and $C_2H_5NHC_2H_5$(D).

**Options :**

6952787734. $C < B < A < D$

6952787735. $D < C < B < A$

6952787736. $C < D < B < A$

6952787737. $D < B < A < C$

**Question Number : 69 Question Id : 6952782204 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match the **LIST-I** with **LIST-II**

| **List-I Deficiency Disease** | | **List-II Vitamin** | |
|---|---|---|---|
| A. | Scurvy | I. | Pyridoxine |
| B. | Convulsions | II. | Vitamin A |
| C. | Cheilosis | III. | Ascorbic Acid |
| D. | Xerophthalmia | IV. | Riboflavin |

Choose the ***correct*** answer from the options given below:

**Options :**

6952787738. A-I, B-III, C-II, D-IV

6952787739. A-I, B-III, C-IV, D-II

6952787740. A-III, B-I, C-IV, D-II

6952787741. A-III, B-I, C-II, D-IV

**Question Number : 70 Question Id : 6952782205 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match the **LIST-I** with **LIST-II**

| **List-I Amino acid** | | **List-II Positive reaction/Test for functional group present in side chain of amino acid** | |
|---|---|---|---|
| A. | Glutamine | I. | Hinsberg's test |
| B. | Lysine | II. | Neutral $FeCl_3$ test |
| C. | Tyrosine | III. | Ceric ammonium nitrate test |
| D. | Serine | IV. | Hoffman bromamide degradation |

Choose the ***correct*** answer from the options given below:

**Options :**

6952787742. A-IV, B-II, C-I, D-III

6952787743. A-IV, B-I, C-II, D-III

6952787744. A-III, B-II, C-I, D-IV

6952787745. A-IV, B-I, C-III, D-II

# Chemistry Section B

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| **Section Id :** | 695278156 |
| **Section Number :** | 6 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 5 |
| **Number of Questions to be attempted :** | 5 |
| **Section Marks :** | 20 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Sub-Section Id :** | 695278156 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

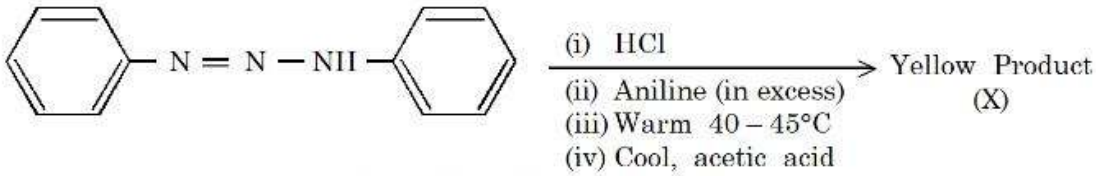
**Question Number : 71 Question Id : 6952782206 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

First and second ionization enthalpies of lithium are 520 kJ $mol^{-1}$ and 7297 kJ $mol^{-1}$ respectively. Energy required to convert 3.5 mg lithium (g) into $Li^{2+}(g)$ [$Li(g) \rightarrow Li^{2+}(g)$] is _____ kJ $mol^{-1}$ . (nearest integer)
[Molar mass of Li = 7 g $mol^{-1}$]

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 72 Question Id : 6952782207 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Consider the following sequence of reactions.




The percentage of nitrogen in the yellow product (X) formed is _____%.
(Nearest Integer)
(Given Molar mass in g $mol^{-1}$ H:1, C:12, N:14)

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 73 Question Id : 6952782208 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

4.7 g of phenol is heated with Zn to give product X. If this reaction goes to 60% completion then the number of moles of compound X formed will be
____ $\times 10^{-2}$.(Nearest Integer)
(Given molar mass in g $mol^{-1}$ : H:1, C:12, O:16)

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 74 Question Id : 6952782209 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Sucrose hydrolyses in acidic medium into glucose and fructose by first order rate law with $t_{1/2} = 3$ hour. The percentage of sucrose remaining after 6 hours is _____. (Nearest integer)
(Given : log 2 = 0.3010 and log 3 = 0.4771)

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 75 Question Id : 6952782210 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Consider the reaction $X \rightleftharpoons Y$ at 300 K. If $\Delta H^{\theta}$ and K are 28.40 kJ $mol^{-1}$ and $1.8 \times 10^{-7}$ at the same temperature, then the magnitude of $\Delta S^{\theta}$ for the reaction in J $K^{-1}$ $mol^{-1}$ is_____. (Nearest integer)
(Given : R = 8.3 J $K^{-1}$ $mol^{-1}$, ln 10 = 2.3, log 3 = 0.48, log 2 = 0.30)

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1